

DHARAVATH NAGARAJU

Embedded Systems Engineer | Edge AI & IoT

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PROFILE

Embedded systems engineer (M.S. in progress, Paris) with hands-on hardware-software design experience: CNN inference deployed on target devices (Raspberry Pi 5, Arduino Nano 33 BLE Sense), custom UART protocols, and sensor integration over I2C, SPI and BLE. Comfortable across the full stack: C/C++ firmware, signal acquisition and edge AI deployment. Electrical engineering background (33/11 kV power distribution). Seeking an apprenticeship or role in embedded systems, edge AI and IoT.

SKILLS

Embedded Systems: C/C++, firmware development, microcontroller programming, Arduino, Raspberry Pi, real-time systems, IoT, sensor integration

Embedded AI / Vision: TensorFlow Lite, CNN inference, Edge AI, MediaPipe, OpenCV, gesture recognition

Protocols: I2C, SPI, UART (custom hexadecimal frame protocols), BLE, PWM

Programming & Tools: Python, PySerial, Flask, Linux, Git, PlatformIO, Arduino IDE, VS Code, MATLAB, HTML/CSS/JavaScript

Electrical Engineering: 33/11 kV distribution, transformers, relay protection, substation maintenance

PROJECTS

AI Hand Gesture Recognition – CNN, MediaPipe, Raspberry Pi 5, UART

Real-time gesture classification from a live camera feed: CNN trained on MediaPipe hand landmarks, inference deployed on Raspberry Pi 5. Hardware control via a custom hexadecimal UART frame protocol (start byte, command, XOR checksum). Code: github.com/dnagaraju271/Gesture-Recognition-CNN

Magic Wand – Edge AI Gesture Recognition – TensorFlow Lite, Arduino Nano 33 BLE Sense

Fully on-device inference with no cloud dependency: TensorFlow Lite CNN classifying IMU data (I2C) in real time, with BLE output and RGB LED feedback.

Home Monitoring and Sensor Dashboard – IoT, I2C/SPI, UART

End-to-end IoT pipeline: temperature, humidity and dust sensors (I2C/SPI), continuous UART acquisition, live-updating responsive web dashboard (HTML/CSS/JS).

Arduino Radar and Stepper Motor Control – PWM, Ultrasonic Sensing

Object detection with about 95% accuracy (2–100 cm range, response under 1 s). Reduced speed fluctuations by 20–25% and held 1.8° step accuracy through optimised PWM control logic.

UPS Battery Monitoring with Automatic Switchover – Relay Control

Cut power interruption time by about 90% using automatic relay switching combined with real-time voltage monitoring.

PROFESSIONAL EXPERIENCE

Process Associate – Genpact

Jul 2024 – Aug 2025

Reduced operational errors by about 15% through systematic double-checks on 250+ daily cases across 10+ SOPs.

Maintained 98–99% processing accuracy and 100% compliance in quality, security and reporting audits.

Intern, Field Electrical Engineer – SPDCL, Telangana, India

Jun 2023 – Aug 2023

Supported field operations on 33/11 kV distribution networks and 10+ transformer maintenance cycles, in strict compliance with electrical safety protocols.

Intern, Electrical Systems – AVR Power Suppliers, Hyderabad, India

Jun 2020 – Dec 2020

Learned transformer design (step-up, step-down, isolation) and relay protection systems (overload, earth fault, differential) across the power transmission chain.

EDUCATION

M.S. Embedded Systems – Paris Institute of Digital Technology (ISEP), France

2025 – 2027

B.Tech Electrical and Electronics Engineering – JNTUH, India (CGPA: 7.8/10)

2021 – 2024

Diploma in Electrical and Electronics Engineering – SBTET, India (CGPA: 8.45/10)

2018 – 2021

LANGUAGES

English: C2 | French: A2